RAG Test Harness

Empirical evaluation, automated self-refining sweeps, and regression detection for vector search.

Soumitra Ghosh
IIT KGP 1996 B.Tech. ECE

CTO, Knio Infotech
Enterprise AI & Retrieval Architecture

PROBLEM STATEMENT & OBJECTIVE

The Goal: Accelerating RAG to Production

Bridging the gap between empirical evaluation and high-velocity engineering.

session@rag-test-harness: ~/manifesto.sh

```
$ cat << 'STATEMENT'

In the post-LLM era, traditional searches are to be replaced by LLM backed RAG systems. Engineers will spend numerous hours implementing and optimizing RAG projects. Let's explore how AI can help improve key metrics while reducing the time to market for such systems.

STATEMENT
```

Core Mandate: Replace subjective guesswork with rigorous AI-driven evaluation sweeps, turning RAG deployment from months of trial-and-error into automated, measurable engineering.

FOUNDATIONS · CORE CONCEPTS

Why RAG: Grounding Enterprise Intelligence

Foundation models offer extraordinary reasoning, but lack private enterprise context, real-time awareness, and verifiable citations.

01 · HALLUCINATION SHIELD

Mitigate Hallucination

Parametric memory in LLMs confidently fabricates plausible facts. RAG anchors generation to verifiable retrieved source chunks.

Enterprise Impact

Eliminates high-risk confabulations in legal, clinical, and compliance tasks.

`Advantage: Drastic drop in factual drift`

02 · PRIVATE & DYNAMIC DATA

Private & Fresh Knowledge

Model pretraining cuts off at fixed dates. RAG connects enterprise vector databases, APIs, and live internal documents dynamically.

Zero Retraining Lag

Instant updates when documents change without costly fine-tuning pipelines.

`Advantage: Real-time sync with sources`

03 · AUDITABILITY & TRUST

Verifiable Citations

Black-box outputs prevent governance sign-off. RAG guarantees exact sentence-to-chunk provenance for human review and auditing.

Governance & Auditing

Complete audit trails for regulatory compliance, traceability, and enterprise trust.

`Advantage: 100% source-attributed answers`

RAG converts probabilistic generative models into verifiable, source-grounded reasoning engines—making enterprise AI trustworthy, compliant, and continuously updated.

FOUNDATIONS · CORE CONCEPTS

RAG Fundamentals: The Three Pillars

01 · INGESTION & INDEX

Chunking & Embedding

Raw enterprise corpora are partitioned into semantic windows and mapped to dense dense vector spaces.

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Key knob: chunk size, overlap, embedding model dimension
```

Foundation for all recall

02 · RETRIEVAL & RANK

Hybrid Search & Rerank

Vector similarity combined with BM25 keyword matching, refined via cross-encoder rerankers.

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Key knob: Top-K candidates, score thresholds, cross-attention, multi-hop graph traversal
```

Where 70%+ failures originate

03 · GROUNDED SYNTHESIS

Cited Generation

LLM crafts factually bound answers strictly conditioned on the retrieved passages with explicit source citations.

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Key knob: system prompt, citation parser, hallucination guard
```

Output contract with user

The Central Challenge: Each pillar contains 5+ tunable hyperparameters. Interdependent variables create explosive combinatorial search spaces.

THE PROBLEM STATEMENT · ENGINEERING REALITIES

Challenges I faced

01 · COMBINATORIAL SEARCH
Tuning & Corpus

- 15+ Fine-Tuning Parameters
- Corpus Variations
- _

02 · GRAPH & INGESTION
Graph & Representation

- Corpus-Dependent Graph
- Dimensionality of Chunking
- Provider Abstraction

03 · EVALUATION & QUALITY
Evaluation & Gates

- Comprehensive Metrics
- Golden QA Bootstrapping
- Automated Quality Gates

04 · INFRASTRUCTURE
Runtime & Cost

- Continuous Monitoring
- Ollama Load Balancing
- Long Gaps Between Phases

Manual experimentation cannot scale across 15+ variables and shifting corpora.
A systematic test harness transforms ad-hoc guesswork into automated, empirical engineering.

HARNESS ENGINE · SUBSYSTEM ARCHITECTURE

Test Harness Architecture



01 · DATASET & EVAL SUITE

Golden Evaluation Suite

Code Generator

02 · EXECUTION ENGINE

Iterative Sweep Runner

Budget aware, HITL workflow

03 · TELEMETRY & RCA

Telemetry & RCA Agents

Architectural Invariant: All evaluation runs are strictly deterministic, isolated from production databases, and budget-gated.

Summary — Report Card

Evaluating target deliverables, AI-generated infrastructure, and autonomous harness maturity against set objectives.

PIPELINE AUTOMATION
Autonomous Execution

90% Automated Sweeps
[ACHIEVED]

Ingestion & Eval Pipelines
[ACHIEVED]

Retriggering Failed Evals
[ACHIEVED]

Restarting Paused Sweeps
[ACHIEVED]

AI CODE GEN & QUALITY
Synthesis & Gates

Code Gen: OCR & Graph Modules [ACHIEVED (90%)]

Golden QnA Generation
[ACHIEVED]

Quality Gate Enforcement
[ACHIEVED]

GOVERNANCE & CONTROLS
Budget & Convergence

Sweep Budget Guardrails
[ACHIEVED]

Convergence Detection
[ACHIEVED (80%)]

Overall Goal Attainment [85% COMPLETED]

Report Card Verdict: Autonomous sweep exploration and foundational pipelines reached 90%+ target completion; deterministic quality gates and budget controls strictly enforced.

ENGINEERING PROCESS

Building the Harness with AI

THE PROCESS

Specs Before Code: Authored rigid architecture blueprints and interface boundaries first.

Phased Delivery: Built data layer, eval runner, then autonomous agent loop sequentially.

Review-Fix Commits: Forced AI to diff its own changes against explicit criteria before merging.

Real-DB Coverage: Zero mocks. Verified with real vector indices and live SQL databases.

Disciplined software engineering applied to generative LLM coding.

DECISIONS I MADE & REJECTED

Read-Only SQL for RCA: Rejected arbitrary DB write access; prevented hallucinations from mutating state.

Budget Guardrails: AI agent wanted infinite sweeps. Hard-capped spending at $5 per run.

Reranker Behind Switch: Avoided mandatory deployment since cross-encoders spike latency 300%.

Human oversight determined architectural sanity and cost control.

"AI wrote the boilerplate at 10x speed. I had to decide the trust boundaries."

OPEN DISCUSSION · INTERACTIVE SESSION

Questions & Technical Discussion

Deep dive into retrieval architectures, automated sweeps, failure modes, and production trade-offs.



Floor Open for Inquiries · Edge-cases & production sweeps

STARTER PROMPTS & KEY THEMES

Frequently Raised Discussion Topics

Q1 · EMBEDDINGS & RETRIEVAL

How do you systematically detect and resolve retrieval regressions when swapping dense embedding models or tuning chunk overlap?

Context: Metric drops on specific enterprise document slices.

Q2 · SCALING & GRAPH RAG

What is the real-world latency and cost trade-off between standard hybrid Vector+BM25 search vs. multi-hop Graph RAG architectures?

Context: Production SLAs, indexing cost, and entity community summaries.

Q3 · GROUNDEDNESS & EVAL HARNESS

Can LLM-as-a-judge (RAGAS / TruLens) run reliably in automated CI/CD sweeps without suffering from evaluation bias and variance?

Context: Reference-free faithfulness scoring and synthetic golden test sets.

Soumitra Ghosh · github.com/soumitra74